Practice Set 5 Paper 2 – Urology

1. Q1. Describe metabolic evaluation in stone disease
2. Q2. Describe principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD
3. Q3a. Bladder preserving approach in invasive TCC of bladder
4. Q3b. mRCC.
5. Q3c. Dexamethasone suppression test
6. Q3d. HIFU
7. Q3e. Discuss techniques of dialysis
8. Q3f. RIRS

Practice paper compiled from the question bank.

Q1. Describe metabolic evaluation in stone disease

Type: LAQ (25 marks) (also answered in 2015-Summer Paper 2, Q2 and 2025-Winter Paper 1, Q3b)

Introduction and rationale

- Stone disease is a chronic metabolic disorder: first-time formers have about a **50% risk of recurrence within 10 years**.
- Evaluation finds the treatable abnormality; therapy matched to 24-hour urine results reduces stone events. The **ROKS (Recurrence of Kidney Stone) nomogram** estimates individual risk.

Physicochemical basis (why we measure these)

- Stones form when urine is **supersaturated**: nucleation, growth, aggregation and retention, often on **Randall plaque**.
- In the **metastable zone** (between solubility and formation products) crystals grow on existing nuclei but do not form spontaneously.
- **Promoters** (calcium, oxalate, uric acid, low volume) raise supersaturation; **inhibitors** (citrate, magnesium) oppose crystallisation. The 24-hour urine measures both.

Step 1: Screening evaluation (every first-time former; AUA [American Urological Association] and EAU [European Association of Urology])

Component	What is assessed
History	GI (gastrointestinal) disease or bariatric surgery, gout, sarcoidosis, recurrent UTI (urinary tract infection), family history; diet; lithogenic drugs: calcium, vitamins C and D, acetazolamide, topiramate, steroids, triamterene
Blood	Creatinine, sodium, potassium, calcium, uric acid, bicarbonate; **PTH (parathyroid hormone)** if calcium is high or high-normal
Urine	pH, crystals (hexagonal = cystine), culture; cyanide–nitroprusside cystine screen
Stone analysis	Infrared spectroscopy or X-ray diffraction: uric acid points to metabolic syndrome, cystine to a genetic cause, struvite to infection
Imaging	NCCT (non-contrast computed tomography): burden predicts future events; uric acid usually <500 HU (Hounsfield units)

- Screening is a **triage**: low risk gets general advice; high risk goes on to 24-hour urine.

Who is high risk

- Recurrent formers, **children**, solitary kidney, bilateral or large burden, GI disease, obesity, metabolic syndrome, diabetes, primary hyperparathyroidism, RTA (renal tubular acidosis), sarcoidosis, gout, family history; uric acid, cystine or infection stones.

Step 2: Extended metabolic evaluation

- **One or two 24-hour urines** on the usual diet, at least 4 weeks after the episode, stone-free and infection-free. Creatinine checks completeness (men about 20–25, women about 15–20 mg/kg/day).

Parameter	Desirable per day	Abnormality and meaning
Volume	**>2.5 L**	Low volume: commonest, easiest to fix
Calcium	<250 mg women, <300 mg men (or <4 mg/kg)	Hypercalciuria

Parameter	Desirable per day	Abnormality and meaning
Oxalate	**<40 mg**	Dietary, **enteric** (fat binds calcium, freeing oxalate), **primary** (often >100 mg)
Citrate	**>320 mg**	Hypocitraturia: distal RTA, diarrhoea, hypokalaemia, animal protein
Uric acid	<600 mg	Hyperuricosuria promotes calcium oxalate stones
pH	5.8–6.2	<5.5 uric acid; >6.5 calcium phosphate, RTA, infection
Sodium	<100 mmol	High sodium raises calcium excretion
Cystine	Negative screen	Cystinuria; keep <250 mg/L

- **Also know:** magnesium, potassium, phosphate; RSS (relative supersaturation) is calculated from these.

Interpreting hypercalciuria (Pak classification)

Type	Defect	Serum calcium	PTH	Fasting urine calcium
Absorptive (20–40%)	Increased gut absorption	Normal	Normal or suppressed	Normal
Renal leak (5–8%)	Tubular calcium loss	Normal	High (secondary)	High
Resorptive (3–5%)	Primary hyperparathyroidism	High	High	High

- Fast and calcium-load subtyping is no longer advised, because it does not change treatment.

Special tests

- **Distal RTA:** urine pH >6, low citrate, calcium phosphate stones. Incomplete form: **ammonium chloride load** (0.1 g/kg) or furosemide–fludrocortisone test fails to lower urine pH below 5.5.
- **Cystinuria:** quantitative 24-hour cystine; solubility about 300 mg/L at pH 5 and 400 mg/L at pH 7.
- **Primary hyperoxaluria:** genetic testing (type 1, AGXT gene, about 70%).
- **Also know:** medullary sponge kidney; calyceal diverticulum; 2,8-dihydroxyadenine stones.

Linking results to treatment

Abnormality	Treatment
Low volume	Fluid for urine >2.5 L/day
Hypercalciuria	Thiazide (hydrochlorothiazide 25 mg twice daily) plus sodium restriction; parathyroidectomy if primary hyperparathyroidism
Hypocitraturia, distal RTA, uric acid stones	Potassium citrate (urine pH 6.0–6.5 for uric acid)
Enteric hyperoxaluria	Low fat and oxalate; calcium with meals
Hyperuricosuria	Allopurinol 300 mg/day
Cystinuria	Urine ≥3 L/day, pH 7.0–7.5, tiopronin

- **Borghi 1996:** urine >2 L/day cut 5-year recurrence (12% vs 27%).

Special groups and follow-up

- **Children:** all evaluated (spot ratios if not toilet-trained). **Pregnancy:** defer 24-hour urine (physiological hypercalciuria).
- **Follow-up (AUA):** repeat 24-hour urine **within 6 months** of starting therapy, then yearly; blood tests for drug effects (potassium on thiazide or citrate); analyse every new stone.

Recent advances

- **Lumasiran** (RNA interference) for primary hyperoxaluria type 1; **NOSTONE (2023):** hydrochlorothiazide did not reduce recurrence.

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Stone episode → screening evaluation
  Low risk → general advice
  High risk → 24-h urine ×1–2 → targeted therapy → repeat ≤6 months → yearly
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Diagram to draw: supersaturation zones (undersaturated, metastable, unstable) and the evaluation algorithm.

Key points to remember:

- Screening for every first-time former; 24-hour urine for high-risk and recurrent formers.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
- Distal RTA: ammonium chloride fails to lower pH below 5.5.
- Borghi 1996: fluid halves recurrence; repeat 24-hour urine within 6 months.

Q2. Describe principles of vascular anastomosis. Enumerate and describe options for vascular access surgery for HD

Type: LAQ (25 marks) (vascular access also answered in 2020-Summer Paper 4, Q3b and 2025-Winter Paper 2, Q3c)

Assumed meaning: HD = haemodialysis.

Introduction

- A vascular anastomosis joins two vessels so that flow is smooth, leak-free and lasting (transplantation, autotransplantation, caval repair, dialysis access).
- **Types:** end-to-end, **end-to-side** (commonest: transplant, fistula), side-to-side, and patch (Carrel aortic patch).

Principles of vascular anastomosis

Principle	How and why
Exposure and control	Proximal and distal control (vessel loops, Rummel tourniquets, bulldog or Satinsky clamps)
Anticoagulation	Systemic heparin before clamping; heparinised saline flush; prevents thrombosis in the static column
Atraumatic handling	Hold the adventitia, never the intima; clamp clear of atheroma, because clamp injury causes intimal dissection and stenosis
Healthy ends	Trim diseased ends; resect venous valve cusps at the suture line
No tension, twist or kink	Adequate length; trial the final position before suturing
Intima-to-intima eversion	Smooth endothelium faces the lumen; adventitia and rough surfaces stay outside, as they are thrombogenic
Wide opening	Spatulate or bevel small vessels, or use end-to-side with an elliptical arteriotomy (aortic punch), to avoid stenosis
Suture	**Non-absorbable monofilament polypropylene**, double-armed: smooth, low drag, low thrombogenicity. IVC (inferior vena cava) or aorta 3-0/4-0; transplant vessels 5-0/6-0; wrist fistula about 6-0/7-0
Needle direction	Equal, small, close bites; pass **inside-out on the artery** so plaque and intima are tacked down, not lifted into a flap
Continuous or interrupted	Continuous is quick and haemostatic; **interrupted** for small vessels and children, allowing growth
Technique	**Carrel triangulation** (three stay sutures keep the back wall clear); **parachute** (sew loosely, then draw up; one knot) or tack both apices and run to the midpoint
Flush and release	Before the last knot, briefly release clamps to flush air, clot and debris; leave slack ("growth factor") to avoid purse-stringing; release venous/distal clamp first, then arterial
Confirm	Haemostasis, pulse, thrill, Doppler

- **Failure:** early from technical error (thrombosis, leak, stenosis); late from **intimal hyperplasia**.
- **Also know:** vascular staplers; grafts (autologous vein for small or contaminated fields; PTFE [polytetrafluoroethylene] for large vessels).

Options for vascular access surgery for HD

- Haemodialysis needs a blood flow of about **250–350 mL/min**.

Option	Construction	Ready	Comment
1. Non-tunnelled CVC (central venous catheter)	Dual-lumen 10–12 Fr in internal jugular or femoral vein	At once	AKI (acute kidney injury) and emergencies; days only
2. Tunnelled cuffed CVC	Right internal jugular; cuff blocks bacterial migration	At once	Bridge or last resort; infection, central stenosis
3. Native AVF (arteriovenous fistula)	Artery joined to a superficial vein, usually vein end to artery side	6–8 weeks	**First choice:** best patency, fewest infections
a. Snuffbox or **radiocephalic (Brescia–Cimino)**	Radial artery to cephalic vein at wrist		Most distal; preserves proximal sites; higher failure to mature
b. Brachiocephalic	Brachial artery to cephalic vein at elbow		Higher flow; more steal
c. Brachiobasilic transposition	Deep basilic vein mobilised and moved superficially	Longer, often two stages	When the cephalic vein is unusable
4. AVG (arteriovenous graft)	PTFE forearm loop, upper-arm or thigh	2–3 weeks	Poor veins; more stenosis (venous anastomosis), thrombosis, infection
5. HeRO (Haemodialysis Reliable Outflow) graft	Graft outflow bypasses occluded central veins into the right atrium		Central vein occlusion
6. Endovascular AVF	Percutaneous radiofrequency or thermal anastomosis between adjacent forearm vessels		No incision; selected anatomy

- **Also know:** lower-limb femoral vein transposition when arms are exhausted; peritoneal dialysis needs no vascular access.

Planning

- **Hierarchy:** AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first**; individualised (KDOQI [Kidney Disease Outcomes Quality Initiative] 2019 "Life-Plan").
- **Timing:** refer at eGFR (estimated glomerular filtration rate) <30; create the AVF about **6 months** before dialysis.
- **Vein preservation:** no venepuncture or PICC (peripherally inserted central catheter) in that arm; avoid subclavian lines.
- **Assessment:** pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm).
- **Maturity "rule of 6s":** flow >600 mL/min, diameter >6 mm, depth <6 mm, at 6 weeks.

Complications of access

Complication	Treatment
Failure to mature; stenosis (neointimal hyperplasia)	PTA (percutaneous transluminal angioplasty), ligate side branches, stent graft
Thrombosis (usually on a stenosis)	Thrombectomy, then treat the stenosis
Steal syndrome (hand ischaemia)	Banding, DRIL (distal revascularisation–interval ligation)
Aneurysm; infection (CVC > AVG > AVF)	Repair; antibiotics, excise graft, remove catheter

- **Also know:** venous hypertension; high-output failure; monitor thrill and bruit.

Diagram to draw: radiocephalic end-to-side fistula with Carrel triangulation stay sutures.

Key points to remember:

- Intima-to-intima eversion, monofilament polypropylene, inside-out on the artery, flush before tying.
- AVF first, distal, non-dominant arm, about 6 months ahead.
- Rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis sits on a stenosis; steal → banding or DRIL.

Q3a. Bladder preserving approach in invasive TCC of bladder

Type: SAQ (10 marks) (also answered in 2019-Summer Paper 4, Q2)

Introduction

- Standard for MIBC (muscle-invasive bladder cancer; TCC = transitional cell carcinoma) is cisplatin NAC (neoadjuvant chemotherapy) plus RC (radical cystectomy). Bladder preservation suits **unfit** patients and **fit, motivated patients with limited disease**.
- Single modalities fail because they leave the urothelium or occult nodes untreated.

Options

Option	Use
Radical TURBT (transurethral resection of bladder tumour) alone	Not a primary treatment
Partial cystectomy + PLND (pelvic lymph node dissection)	Solitary tumour **<3 cm** in mobile bladder (dome), **no CIS (carcinoma in situ)**, 1–2 cm margin, frozen sections; **urachal adenocarcinoma**
RT (radiotherapy) or chemotherapy alone	Inferior
TMT (trimodality therapy)	Best studied, guideline-accepted

Trimodality therapy

- **Components:** (1) maximal **visibly complete TURBT**; (2) RT to bladder ± nodes; (3) concurrent **radiosensitising chemotherapy**.
- **Selection:** cT2, small solitary tumour, **no hydronephrosis, no CIS**, good bladder function, compliant patient. Only 6–19% of MIBC qualify.
- **Radiosensitisers: mitomycin C + 5-FU (fluorouracil)** (**BC2001:** 2-year locoregional DFS [disease-free survival] 67% vs 48% vs RT alone); cisplatin; gemcitabine.
- **Dose:** 64–66 Gy in 32 fractions or **55 Gy in 20** (non-inferior).

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TURBT → chemoradiation → cystoscopy + biopsy at 3–4 months
  Complete response → surveillance
  Persistent or recurrent invasive → salvage cystectomy
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- **Salvage cystectomy** in **10–30%**: the designed safety net. Non-muscle-invasive (NMIBC) recurrence: TURBT and intravesical BCG (bacillus Calmette–Guérin).

Outcomes

- 5-year CSS (cancer-specific survival) 50–84%. Propensity-matched RC vs TMT: equal 5-year metastasis-free survival (about 75%) and CSS (81% vs 84%). No completed randomised trial (SPARE closed).

Follow-up

- Cystoscopy every 3–4 months for 2 years, then less often, for 10 years; CT (computed tomography) urogram and chest imaging; cytology.

Recent advances

- Checkpoint inhibitors with chemoradiation under trial (S1806, KEYNOTE-992); bladder sparing after complete response to NAC (RETAIN: about half recurred).

Diagram to draw: TMT flowchart (as above).

Key points to remember:

- TMT = complete TURBT + RT + radiosensitiser; T2, no hydronephrosis, no CIS.
- BC2001: mitomycin C/5-FU + RT, DFS 67% vs 48%.
- Salvage cystectomy 10–30%; partial cystectomy for solitary <3 cm dome tumour.

Q3b. mRCC.

Type: SAQ (10 marks) (also answered in 2024-Winter Paper 3, Q3b; immunotherapy in detail in 2024-Winter Paper 2, Q3e)

Assumed meaning: management of metastatic renal cell carcinoma (mRCC).

Assessment

- CT (computed tomography) chest, abdomen and pelvis; bone scan or brain MRI (magnetic resonance imaging) if symptomatic; **biopsy** for histology.
- **IMDC (International Metastatic RCC Database Consortium) risk factors:** Karnofsky <80%; haemoglobin below normal; calcium above ULN (upper limit of normal); neutrophils above ULN; platelets above ULN; <1 year from diagnosis to systemic therapy.
- Favourable 0, intermediate 1–2, poor ≥3; median survival 43.2, 22.5 and 7.8 months. (MSKCC [Memorial Sloan Kettering] model: 5 factors including LDH [lactate dehydrogenase].)

First-line systemic therapy (clear cell)

IMDC risk	Preferred	If IO (immuno-oncology) not possible
Favourable	Pembrolizumab + axitinib; nivolumab + cabozantinib; pembrolizumab + lenvatinib	Sunitinib, pazopanib; surveillance for indolent low-volume disease
Intermediate/poor	Same three, **or nivolumab + ipilimumab**	Cabozantinib, sunitinib, pazopanib

- **Rationale:** VHL (von Hippel–Lindau) loss → HIF (hypoxia-inducible factor) → VEGF (vascular endothelial growth factor), the target of TKIs (tyrosine kinase inhibitors); checkpoint inhibitors release T-cell brakes.

Later lines

- After IO: a VEGF TKI not used before (cabozantinib, axitinib, tivozanib) or **belzutifan** (HIF-2α inhibitor). IO rechallenge is not advised (CONTACT-03 negative).

Cytoreductive nephrectomy (CN)

- **CARMENA (2018):** sunitinib alone non-inferior to CN + sunitinib in intermediate/poor risk. **SURTIME:** deferred CN safe.
- Upfront CN only for fit, low-burden patients; **deferred CN** in responders; avoid in poor risk.

Metastasectomy and radiotherapy

- Complete resection of limited metastases: 5-year survival 35–50%.
- **SBRT (stereotactic body radiotherapy)** for oligometastases; radiosurgery for brain; palliative radiotherapy for bone pain.

Supportive and special groups

- Zoledronic acid or denosumab for bone. Non-clear cell: trials, cabozantinib or sunitinib. Adjuvant pembrolizumab after complete metastasectomy (KEYNOTE-564).

Diagram to draw: flowchart: biopsy + IMDC → IO combination → deferred CN in responders → switch class on progression.

Key points to remember:

- IMDC 6 factors; poor = ≥3.
- IO combinations for all risks; nivolumab + ipilimumab for intermediate/poor.
- CARMENA and SURTIME: no routine upfront CN.
- Belzutifan after IO and TKI.

Q3c. Dexamethasone suppression test

Type: SAQ (10 marks)

Principle

- **Dexamethasone** is a potent synthetic glucocorticoid that is **not detected by the cortisol assay**. In a normal axis it suppresses pituitary ACTH (adrenocorticotropic hormone) by negative feedback, so morning cortisol falls.
- **Autonomous** cortisol (adrenal adenoma) or excess ACTH drive (pituitary, ectopic) fails to suppress.
- "Low-dose" 1 mg is still 3–4 times physiological glucocorticoid.

Indications

- Suspected Cushing syndrome; **every adrenal incidentaloma** (5–8% secrete cortisol); suspected ACC (adrenocortical carcinoma).

Types

Test	Method	Abnormal
Overnight 1-mg LD-DST (low-dose DST; preferred screen)	Dexamethasone 1 mg at 11 pm–midnight; serum cortisol 8–9 am	See below
48-hour 2-mg LD-DST	0.5 mg 6-hourly for 48 hours; cortisol at 48 hours	>1.8 µg/dL; improves accuracy
High-dose DST	8 mg overnight or 2 mg 6-hourly for 48 hours	Over 50% suppression suggests pituitary source; **no longer routine** (some ectopic tumours also suppress)

Interpreting the 1-mg test

Morning cortisol	Meaning
≤1.8 µg/dL (50 nmol/L)	Normal; >90% sensitive, rules out
1.8–5.0 µg/dL	Possible autonomous cortisol secretion
>5.0 µg/dL (140 nmol/L)	Autonomous secretion; >95% specific

- ESE (European Society of Endocrinology) 2023 incidentaloma guideline: cortisol >1.8 µg/dL without overt signs = **MACS (mild autonomous cortisol secretion)**.
- If abnormal: repeat testing and measure **morning ACTH** (low = adrenal; high = pituitary or ectopic, IPSS [inferior petrosal sinus sampling]).

Pitfalls

- **False positive:** oral contraceptives (raise cortisol-binding globulin; up to 50%); CYP3A4 inducers (phenytoin, carbamazepine, rifampicin); missed dose; pseudo-Cushing states (depression, alcohol, obesity, stress).
- **False negative:** CYP3A4 inhibitors (itraconazole, ritonavir, diltiazem).
- **Choice of test:** pregnancy → UFC (urinary free cortisol); epilepsy → LNSC (late-night salivary cortisol); renal failure → DST (not affected by GFR [glomerular filtration rate]).
- Exclude exogenous steroids (including creams and sprays) first.

Urological relevance

- Autonomous cortisol suppresses the opposite adrenal: give **perioperative steroid cover** and taper after adrenalectomy.

Diagram to draw: HPA (hypothalamic–pituitary–adrenal) axis with dexamethasone feedback.

Key points to remember:

- 1 mg at 11 pm; 8 am cortisol ≤1.8 µg/dL excludes; >5.0 µg/dL confirms autonomy.
- Oral contraceptives and enzyme inducers cause false positives.
- High-dose DST no longer routine; ACTH separates adrenal from pituitary/ectopic.

Q3d. HIFU

Type: SAQ (10 marks)

Definition and principle

- HIFU (high-intensity focused ultrasound) focuses ultrasound from a bowl-shaped transducer or lens onto a point. Energy is absorbed only where the beams converge, so tissue there is heated while the path is spared.
- **Intensity 1500–10,000 W/cm²** (diagnostic ultrasound causes negligible heating).
- **Mechanisms: thermal coagulative necrosis** (≥55 °C; prostate exceeds 100 °C within seconds) plus **cavitation** (bubbles collapse and damage cells).
- Each lesion is a **rice grain**; the target is "painted" with overlapping lesions. Ablated and viable tissue look alike on ultrasound, so gaps occur.

Devices and routes

- **Transrectal** probe that both monitors and treats (posterior zone).
- **TULSA (MRI [magnetic resonance imaging]-guided transurethral ultrasound ablation): MR thermometry** confirms lethal temperature; reaches the anterior gland.

Indications

Setting	Use
Focal therapy, prostate cancer	MRI-visible, unilateral, grade group 2–3 index lesion; hemiablation; based on the index-lesion hypothesis
Whole-gland ablation	Localised cancer; retreatment 29–38%
Salvage after EBRT (external beam radiotherapy)	Biopsy-proven local recurrence, **≥2 years after EBRT**
BPH (benign prostatic hyperplasia)	Approved
Kidney	Investigational (incomplete ablation)

Limitations

- **Anterior lesions** (energy lost, oedema displaces target), large gland, gland height and **calcification** (some advise TURP [transurethral resection of the prostate] first).

Outcomes

- **Focal HIFU:** continence 90–95%, potency 50–78%; infield positive biopsy 16.7–54% (csPCa [clinically significant prostate cancer] 0–21.8%).
- **HEAT registry** (1379 men): 7-year failure-free survival 69%.
- **Salvage HIFU:** biochemical progression-free survival about 48% at 3 years; continence 64–87.5%.

Complications

- Retention (about 8%), UTI (urinary tract infection) 6–18%, stricture (4%), ED (erectile dysfunction); rectourethral fistula rare (about 2% after salvage); osteitis pubis.

Follow-up

- PSA (prostate-specific antigen) falls only 50–80%; **mpMRI (multiparametric MRI) + targeted biopsy at 3–6 months**, then at 1–2 and 5 years. EAU (European Association of Urology): focal therapy within trials or registries.

Also know: histotripsy (>10,000 W/cm², mechanical liquefaction, visible bubble cloud).

Diagram to draw: transrectal transducer focusing beams on a rice-grain lesion in the posterior prostate.

Key points to remember:

- Focused ultrasound → ≥55 °C coagulative necrosis plus cavitation.
- Transrectal HIFU suits posterior lesions; TULSA for anterior.
- Focal HIFU: continence 90–95%; HEAT 7-year failure-free 69%.

Q3e. Discuss techniques of dialysis

Type: SAQ (10 marks) (HD principles also answered in 2017-Winter Paper 2, Q1; CAPD in 2018-Summer Paper 2, Q3e)

Assumed meaning: principles and techniques of HD (haemodialysis) and PD (peritoneal dialysis), with continuous therapies.

Physical principles

Process	Driving force	Removes
Diffusion	Concentration gradient	Small solutes (urea, potassium); bicarbonate enters blood
Convection	Transmembrane pressure; solute dragged with water	Middle molecules (β2-microglobulin)
Ultrafiltration	Hydrostatic (HD) or osmotic (PD) gradient	Water

Haemodialysis

- Blood (250–350 mL/min) passes through a hollow-fibre **synthetic membrane**; dialysate runs **countercurrent** to keep a gradient along the dialyser.
- Heparin (citrate if bleeding risk); about 4 hours, 3 times a week.
- **Access:** AVF (arteriovenous fistula), AVG (arteriovenous graft) or CVC (central venous catheter).
- **Adequacy:** Kt/V ≥1.2 (target 1.4); URR (urea reduction ratio) ≥65%.
- **Complications:** hypotension (commonest), cramps, **disequilibrium syndrome**, access infection, thrombosis, steal.

Peritoneal dialysis

- The **peritoneum** is the membrane: diffusion for solutes; **glucose or icodextrin** draws water osmotically.
- **Tenckhoff catheter** (two Dacron cuffs), placed open, laparoscopically (omentopexy) or percutaneously.
- **CAPD (continuous ambulatory PD):** about 4 exchanges of 2 L daily. **APD (automated PD):** night cycler.
- **Adequacy:** weekly Kt/V ≥1.7; PET (peritoneal equilibration test) classifies transport.
- **Complications:** peritonitis, exit-site infection, catheter malfunction, hernia, hyperglycaemia, peritoneal fibrosis.

Continuous and hybrid therapies

- **CRRT (continuous renal replacement therapy):** CVVH (continuous veno-venous haemofiltration; convection), CVVHD (haemodialysis; diffusion), CVVHDF (both). For haemodynamically unstable AKI (acute kidney injury).
- **Also know:** SLED (sustained low-efficiency dialysis); HDF (haemodiafiltration: CONVINCE 2023 lowered mortality versus high-flux HD); home and nocturnal HD.

HD versus PD

	HD	PD
Setting	Centre or home	Home, continuous
Haemodynamics	Rapid shifts	Gentle; preserves residual function
Favoured in	Abdominal surgery, failed PD	Children, cardiac instability, no vascular access

Indications ("AEIOU", refractory to medical treatment)

- Acidosis, Electrolytes (hyperkalaemia), Ingestions (lithium, salicylates), Overload, Uraemia.

Diagram to draw: dialyser with countercurrent flow beside a PD catheter in the pelvis.

> **Key points to remember:**
>
> - Diffusion for small solutes, convection for middle molecules, ultrafiltration for water.
> - HD Kt/V ≥1.2; PD weekly Kt/V ≥1.7.
> - CRRT for unstable AKI; PD preferred in children.

Q3f. RIRS

Type: SAQ (10 marks) (also answered in 2017-Winter Paper 2, Q3e)

Introduction

- RIRS (retrograde intrarenal surgery) is flexible ureterorenoscopy inside the kidney through the natural route. Active deflection, miniaturised digital scopes and the laser made it practical.

Instrument

Feature	Details
Working channel	**3.6 Fr** (laser fibre, basket)
Deflection	≥270° both ways; lower pole angle averages 140° (maximum 175°)
Types	**Fibreoptic** (image degrades); **digital** (CCD [charge-coupled device] chip on tip, better image); **single-use**

Indications

- **Stones:** renal stones **<2 cm**; lower pole <1 cm, or 1–2 cm as an alternative to PCNL (percutaneous nephrolithotomy); patients who must stay on anticoagulants; failed SWL (shock wave lithotripsy); obesity; staged treatment of larger stones.
- **Others:** low-grade UTUC (upper tract urothelial carcinoma); filling defects; calyceal diverticulum; infundibular stenosis; endopyelotomy.

Technique

1. Sterile urine; single-dose antibiotic.
2. Cystoscopy, **safety guidewire**, retrograde pyelogram.
3. **Ureteral access sheath** (often 12/14 Fr): lowers intrarenal pressure and eases repeated passes, but causes ureteric injury in nearly half (severe in about 1 in 8).
4. Low-pressure irrigation. **Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm)** or **thulium fibre laser**: dusting (low energy, high frequency) or fragment-and-basket.
5. Relocate lower pole stones to an upper calyx with a basket.
6. Inspect the ureter on withdrawal. Stent may be omitted if anatomy and function are normal, no injury and no second stage (AUA [American Urological Association]).

- **Unfavourable lower pole:** steep infundibulopelvic angle, long or narrow infundibulum.

Complications

- Sepsis about 5%; stricture about 3%; perforation (a few per cent); bleeding or subcapsular haematoma <1%; avulsion <1%; stent symptoms.

Recent advances

- Single-use digital scopes; thulium fibre laser and pulse-modulated holmium (less retropulsion); **suction and flexible navigable sheaths**; intrarenal pressure sensors; robotic flexible ureteroscopy.

Diagram to draw: flexible scope through an access sheath deflected into the lower pole calyx.

Key points to remember:

- RIRS: stones <2 cm and patients on anticoagulants.
- Access sheath lowers pressure but injures the ureter.
- Sepsis about 5%, stricture about 3%.

Rapid Revision Sheet – Practice Set 5 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- Screening for every first-time former; 24-hour urine for high-risk and recurrent formers.
- Targets: volume >2.5 L, oxalate <40 mg, citrate >320 mg, calcium <250/300 mg.
- Distal RTA: ammonium chloride fails to lower pH below 5.5.
- Borghi 1996: fluid halves recurrence; repeat 24-hour urine within 6 months.

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- Intima-to-intima eversion, monofilament polypropylene, inside-out on the artery, flush before tying.
- AVF first, distal, non-dominant arm, about 6 months ahead.
- Rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis sits on a stenosis; steal → banding or DRIL.

Q3a. Bladder preserving approach in invasive TCC of bladder (SAQ)

- TMT = complete TURBT + RT + radiosensitiser; T2, no hydronephrosis, no CIS.
- BC2001: mitomycin C/5-FU + RT, DFS 67% vs 48%.
- Salvage cystectomy 10–30%; partial cystectomy for solitary <3 cm dome tumour.

Q3b. mRCC. (SAQ)

- IMDC 6 factors; poor = ≥3.
- IO combinations for all risks; nivolumab + ipilimumab for intermediate/poor.
- CARMENA and SURTIME: no routine upfront CN.
- Belzutifan after IO and TKI.

Q3c. Dexamethasone suppression test (SAQ)

- 1 mg at 11 pm; 8 am cortisol ≤1.8 µg/dL excludes; >5.0 µg/dL confirms autonomy.
- Oral contraceptives and enzyme inducers cause false positives.
- High-dose DST no longer routine; ACTH separates adrenal from pituitary/ectopic.

Q3d. HIFU (SAQ)

- Focused ultrasound → ≥55 °C coagulative necrosis plus cavitation.
- Transrectal HIFU suits posterior lesions; TULSA for anterior.
- Focal HIFU: continence 90–95%; HEAT 7-year failure-free 69%.

Q3e. Discuss techniques of dialysis (SAQ)

- Diffusion for small solutes, convection for middle molecules, ultrafiltration for water.
- HD Kt/V ≥1.2; PD weekly Kt/V ≥1.7.
- CRRT for unstable AKI; PD preferred in children.

Q3f. RIRS (SAQ)

- RIRS: stones <2 cm and patients on anticoagulants.
- Access sheath lowers pressure but injures the ureter.
- Sepsis about 5%, stricture about 3%.